Flight Booking Prediction

Data Science Virtual Internship: Strategic Analysis & Predictive Modeling for British Airways

Presented to: Senior Management Team

EXECUTIVE SUMMARY

Data Quality

Explored 50,000 customer records. Identified and resolved 719 duplicate entries to ensure baseline integrity.

Model Selection

Deployed Random Forest Classifier for high interpretability and robust handling of non-linear features.

Business Goal

Optimize marketing spend by identifying "high-intent" customers before they exit the booking funnel.

THE IMBALANCE CHALLENGE

During initial EDA, we observed a massive class imbalance. Only **15%** of sessions resulted in a completed booking.

- **85%** Non-Bookings (Noise)
- **15%** Completed Bookings (Signal)

Standard models tend to ignore the 15% to maximize accuracy, requiring a specialized "Balanced" approach.

DATA PREPARATION WORKFLOW

Process Step	Action Taken	Business Reasoning
Deduplication	Removed 719 duplicate rows	Prevent bias towards repeated sessions
Label Encoding	Converted Days/Channels to Numeric	Format data for Random Forest math logic
Feature Selection	Dropped low-impact metadata	Reduction of model complexity and noise
Class Weighting	Set Weight to 'Balanced'	Force model to prioritize actual buyers

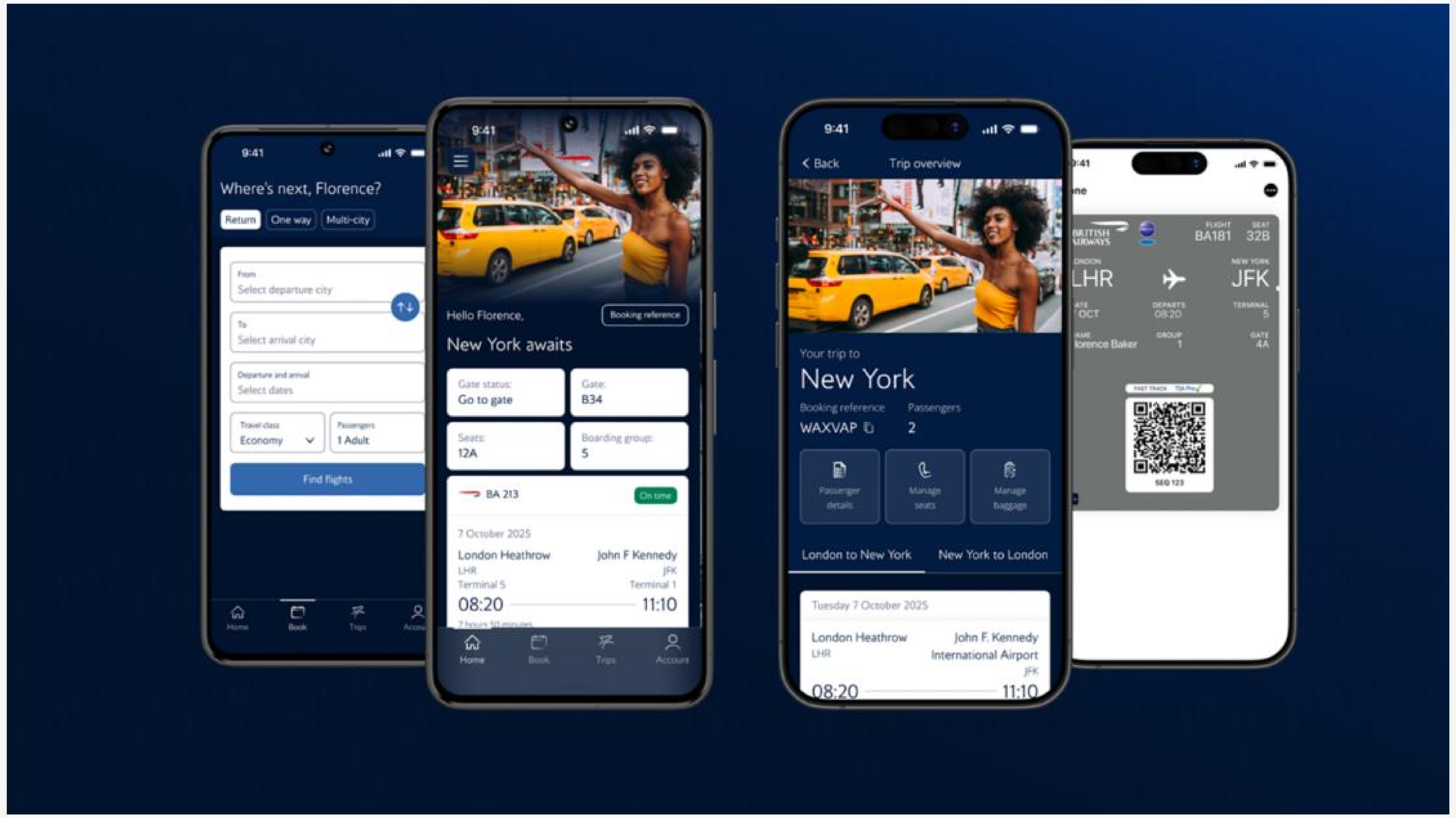
CHANNEL CONVERSION INSIGHTS

Website vs Mobile App

The analysis revealed a critical gap between browsing and buying across different platforms.

- **Internet (Web):** High traffic, established conversion.
- **Mobile App:** Significant traffic but alarmingly low conversion rates.

Insight: The Mobile App UI/UX likely contains friction points preventing checkout completion.



UNMASKING ACCURACY

85%

Raw Accuracy

Why 85% was not enough?

A vanilla model achieved 85% accuracy by simply predicting **everyone** would not book. This is useless for revenue growth.

By applying **Class Balancing**, we shifted focus to the 15% of buyers, sacrificing raw accuracy to gain real **Predictive Recall**.

TOP BOOKING PREDICTORS

The Random Forest model identified these as the top variables influencing a customer's decision:

* Scores represent Feature Importance calculated by the Random Forest algorithm.

Geography: The #1 Predictor

The **Booking Origin** is the most significant factor. Conversion behaviors in regions like Malaysia or Australia differ vastly from others. We must move away from "one-size-fits-all" marketing and implement **region-specific nudges**.

UNDERSTANDING TRIP FRICTION

Flight Duration Effect

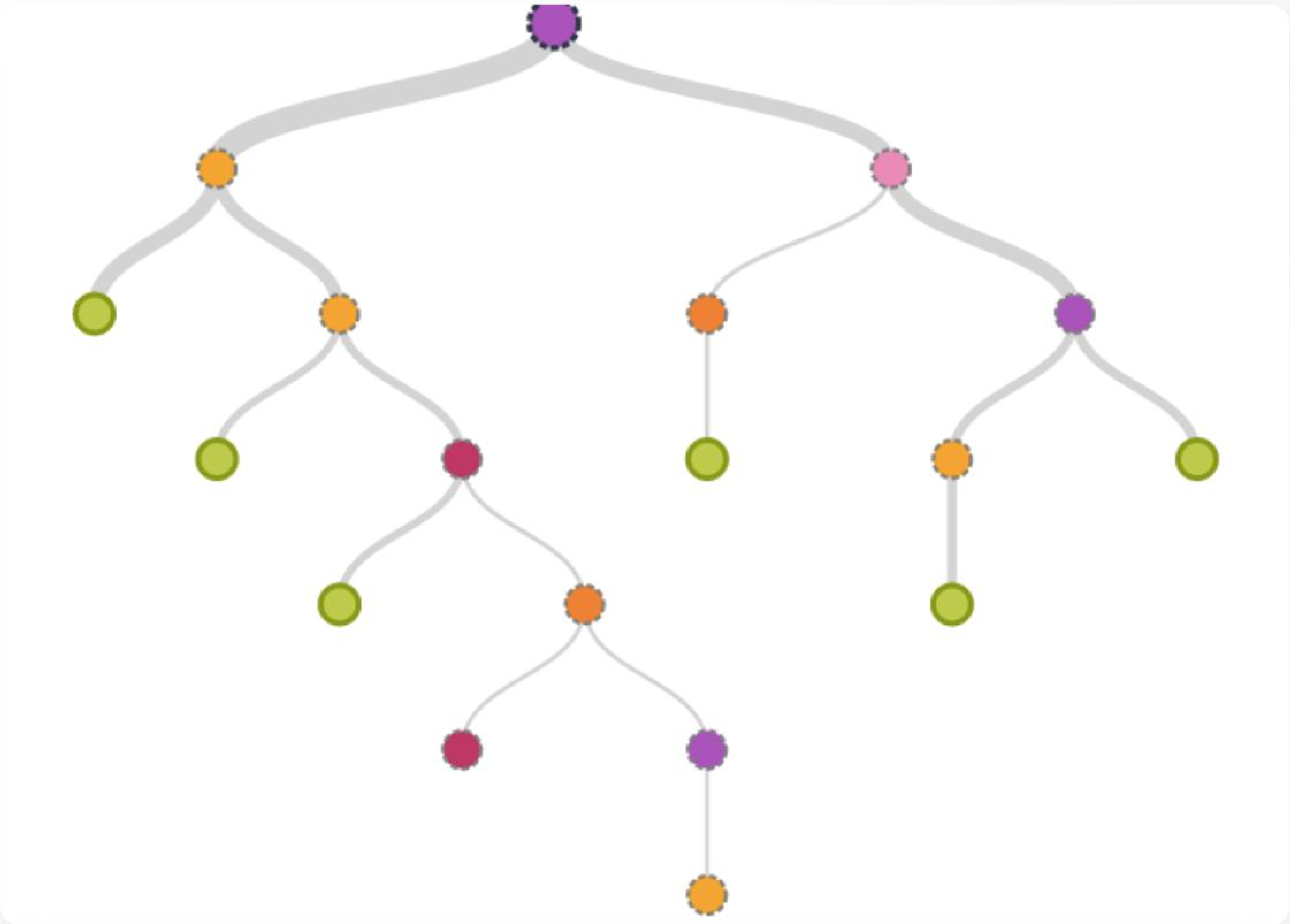
Data shows a strong negative correlation between flight length and booking completion. Long-haul travelers are more prone to price-shopping and session abandonment.

Ancillary Services

Customers who select **Extra Baggage** or **Preferred Seats** are 2.5x more likely to complete a booking. These selections are "Hot Lead" indicators.

RECOMMENDATIONS FOR GROWTH

- **Geo-Targeted Ads:** Focus high-bid marketing on top-converting origins identified by the model.
- **App Revamp:** Audit the mobile booking flow to identify where users drop off.
- **Intent Bundling:** Automatically offer baggage/seat combos to users showing high-intent behaviors.



NEXT STEPS

Phase 1

Deploy real-time scoring API

Phase 2

A/B Test personalized mobile offers

Phase 3

Retrain with Q3/Q4 seasonality

Questions?

Analytical proof provided in the attached Jupyter Notebook.

British Airways Data Science Task - Completed

IMAGE SOURCES

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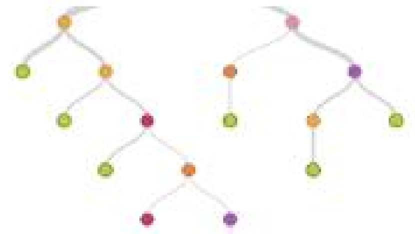

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